Bowling Release Visual Analysis: Tracking and Reconstruction

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2025-2026

Outline

Introduction

- Precise kinematic metrics are fundamental for evaluating player performance in sports
- Ball's rotation dictates the success of a shot
- Commercial systems rely on complex and expensive setups
- Growing demand for systems that work with accessible cameras

Introduction

3D trajectory and spin estimation present challenges

- Uncalibrated camera
- Occlusion, background clutter, changes in illumination
- Varying scene geometries and camera placements

Problem Statement

Estimate the 3D trajectory and spin dynamics of a bowling ball from a static monocular video feed. The pipeline consists of 5 independent tasks:

1. Lane Detection
2. Ball Trajectory Tracking
3. Ball Spin Estimation
4. Planar Metric Rectification
5. 3D Trajectory Reconstruction

State of the Art

Lane Detection and Metric Rectification

- Techniques derived from Advanced Driver Assistance Systems
- Prior knowledge of analyzed scene is leveraged

Ball Tracking and Trajectory Estimation

- Deep learning Frameworks (TrackNet)
- Optical Flow
- Use of event based cameras to capture ultra-fast spin dynamics

Methodology

Implementation of algorithms in *python* for:

- Lane detection
- Ball trajectory tracking
- Ball spin estimation
- Planar metric rectification
- 3D trajectory reconstruction

Methodology: Lane Detection

Bottom Boundary

- Pre-processing
- Edge detection
- Candidate selection

Lateral Boundaries

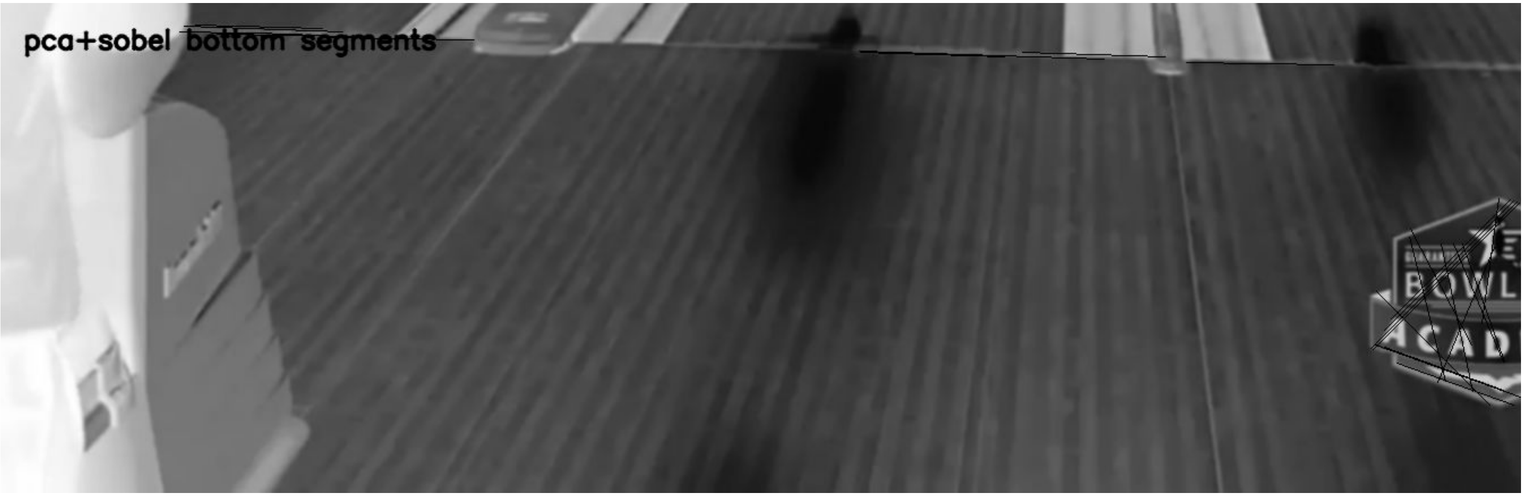
- Pre-processing
- Edge detection
- Candidates selection

Top Boundary

- Pre-processing
- Template matching
- Outlier removal and line fitting

Lane Detection: Bottom Boundary pre-processing

- PCA
- Gaussian blur
- Crop bottom portion



Lane Detection: Bottom Boundary edge detection

- Horizontal Sobel
- Detects Vertical gradients

Lane Detection: Bottom Boundary candidate selection

- HoughLinesP
- Select first (most robust) line that complies with slope threshold

Lane Detection: Lateral Boundaries edge detection

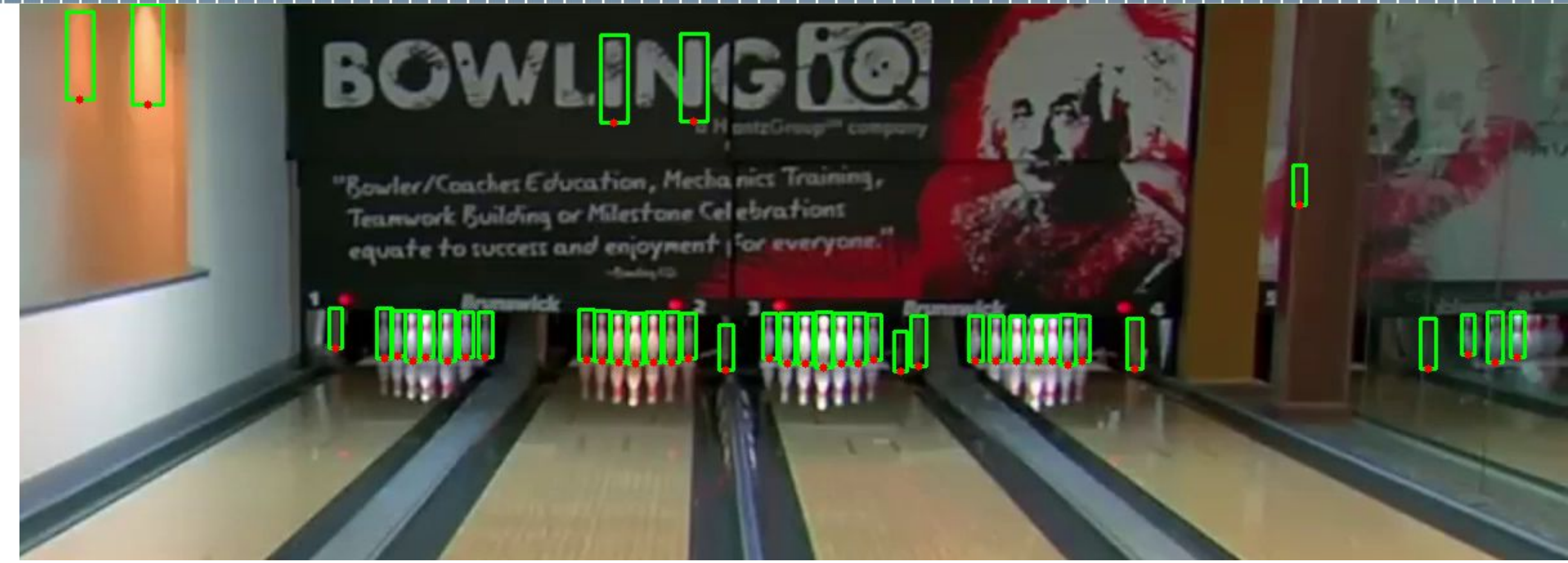
- Vertical Sobel
- Detects horizontal Gradients

Lane Detection: Lateral Boundaries candidates selection

- HoughLinesP
- Select closest lines (horizontal distance) to the left and right of inner lane point

Lane Detection: Top Boundary template matching

- Apply muti scale template matching
- Non maximum suppression
- Select bottom-center coordinates



Lane Detection: Top Boundary outlier removal and line fitting

- Apply RANSAC to detected points to separate inliers from outliers
- Fit order 1 polynomial to inliers

Lane Detection: Final Lane Boundaries

Methodology: Ball Tracking

Trajectory Tracking

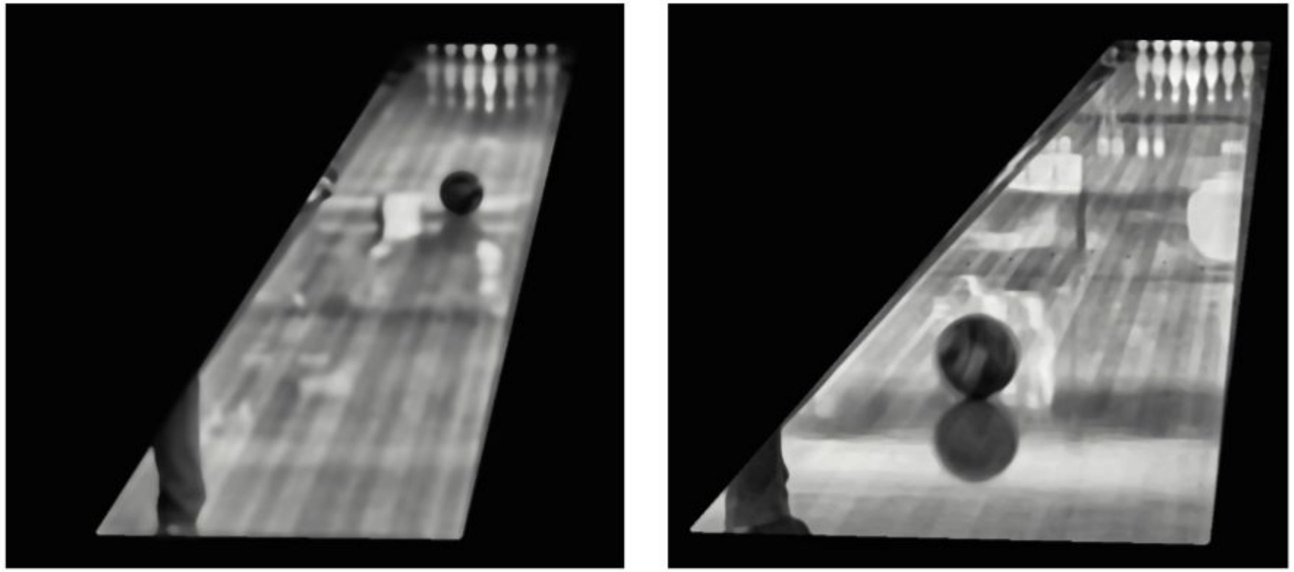
- Pre-processing
- Candidate detection
- Trajectory reconstruction
- Post-processing

Spin Estimation

- Bounding box definition
- Good features retrieval
- Optical flow
- 2D-to-3D Projection
- Axis & angular velocity calculation

Ball Tracking: Trajectory Tracking

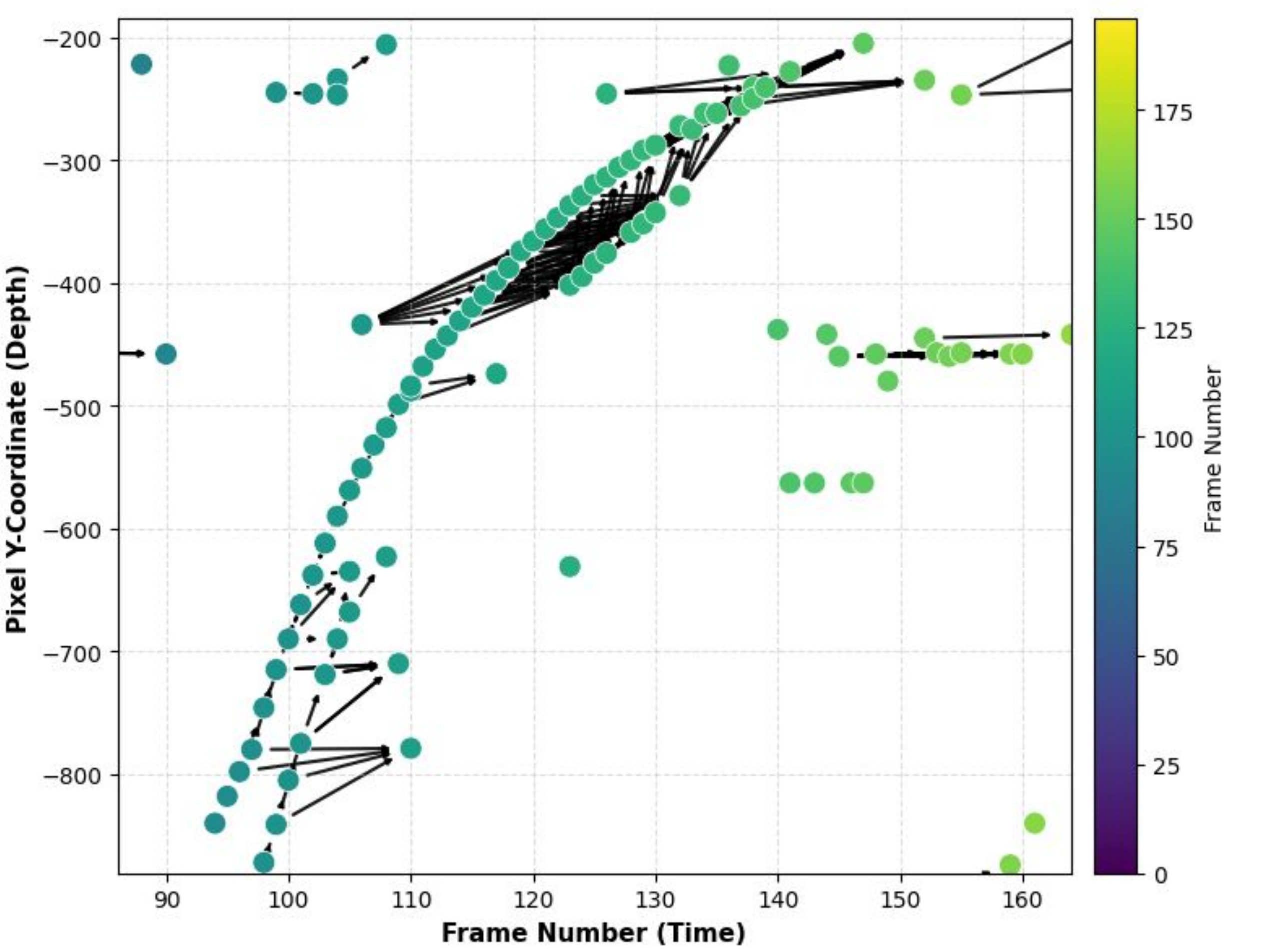
- Pre-processing
  - Lane polygon masking
  - Median blurring to eliminate high frequency noise
  - Localized histogram equalization (CLAHE) to enhance ball contrast



Preprocessed frames of different bowling releases. The ball is clearly distinguishable from the rest of the scene, and the lane imperfections are smoothed out.

Ball Tracking: Trajectory Tracking

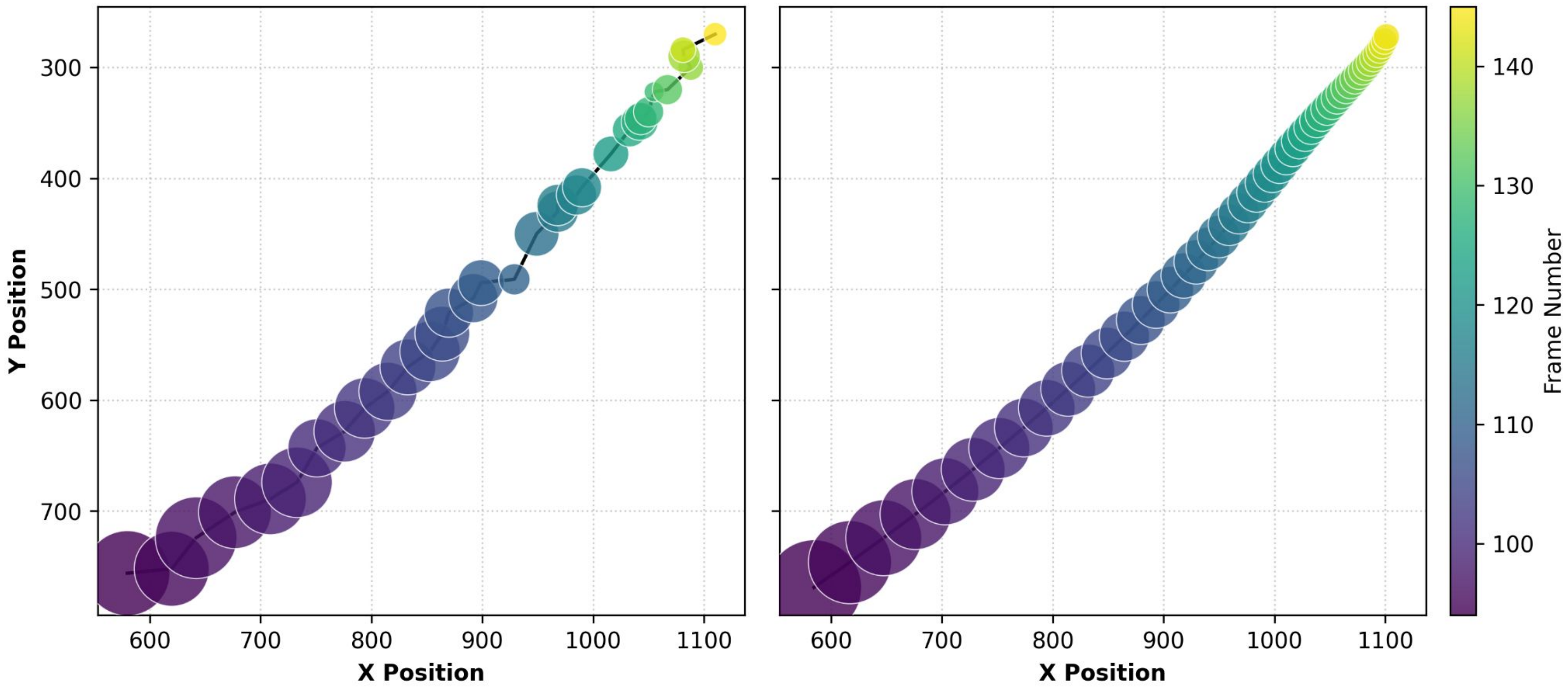
- Candidate detection
  - Built-in OpenCV's Circle Hough Transform
- Trajectory reconstruction
  - Graph representation of the ball candidates across frames
  - Retrieval of the longest path: the most coherent trajectory of the ball



Graph representation of the ball candidates across frames. Each node represents a candidate circle, and directed edges connect candidates in consecutive frames.

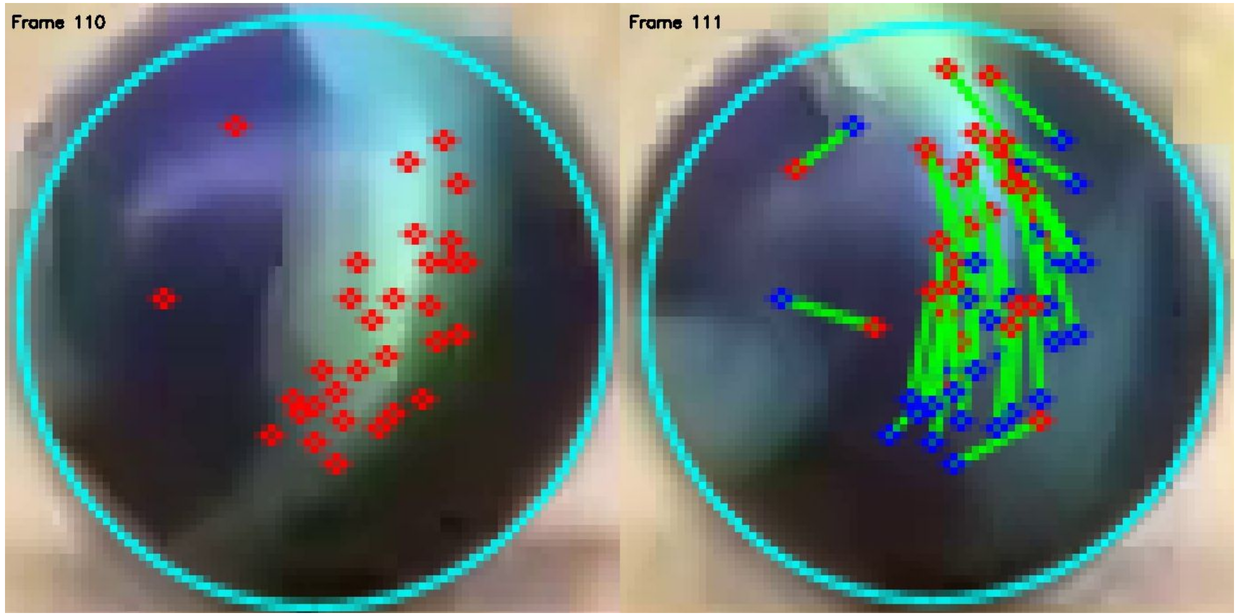
Ball Tracking: Trajectory Tracking

- Post-processing
  - X and Y coordinates, and radii interpolation



Left: Observed ball trajectory, with circle size proportional to observed radius. Right: Final trajectory and radii after interpolation.

Ball Tracking: Spin Estimation



Subsequent frames optical flow. Left: First frame's features detected on the ball's surface. Right: Next frame's feature tracked positions.

- Bounding box definition
- Good features retrieval
  - OpenCVs Shi-Tomasi corner detection algorithm
- Optical flow
  - OpenCVs Pyramidal Lucas-Kanade
  - Forward optical flow
  - Backwards optical flow to filter bad features

Ball Tracking: Spin Estimation

- 2D-to-3D Projection
  - Perfect sphere assumption
- Axis & angular velocity calculation
  - Kabsch algorithm

$$r^2 = X^2 + Y^2 + Z^2$$

$$Z = \sqrt{r^2 - X^2 - Y^2}$$

$$\theta = \arccos\left(\frac{\mathrm{tr}(R) - 1}{2}\right)$$

$$\mathrm{Axis} = \frac{1}{2\sin(\theta)}\begin{bmatrix} R_{32} - R_{23} \\ R_{13} - R_{31} \\ R_{21} - R_{12} \end{bmatrix}$$

$$H = P^T Q$$

$$\mathrm{SVD}(H) = U \Sigma V^T$$

$$R = V U^T$$

Methodology: Lane Rectification and 3D Reconstruction

Planar Metric Rectification

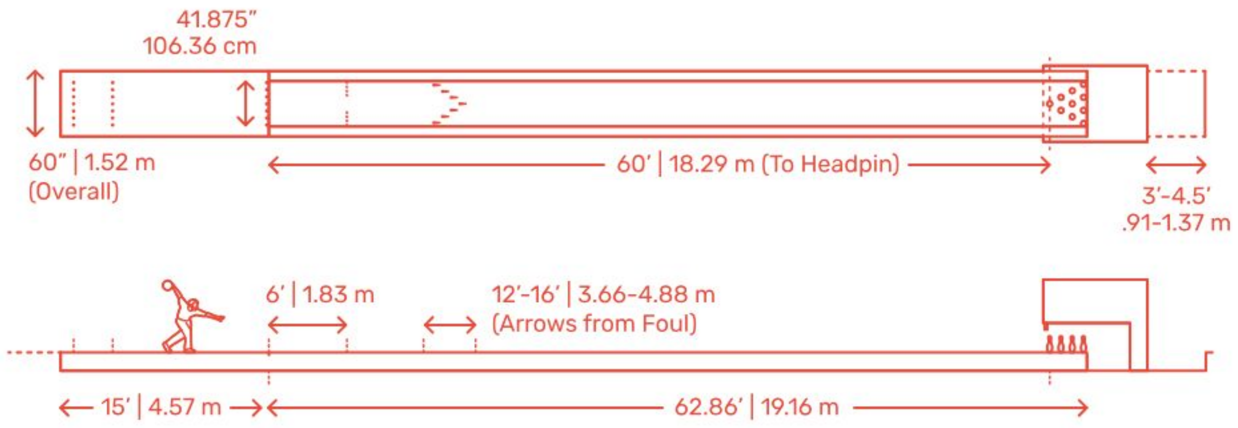
- Homography estimation with known scene geometry

3D Trajectory Reconstruction

- Leverage bowling knowledge for 3D reconstruction

Lane Rectification and 3D Reconstruction: Planar Metric Rectification

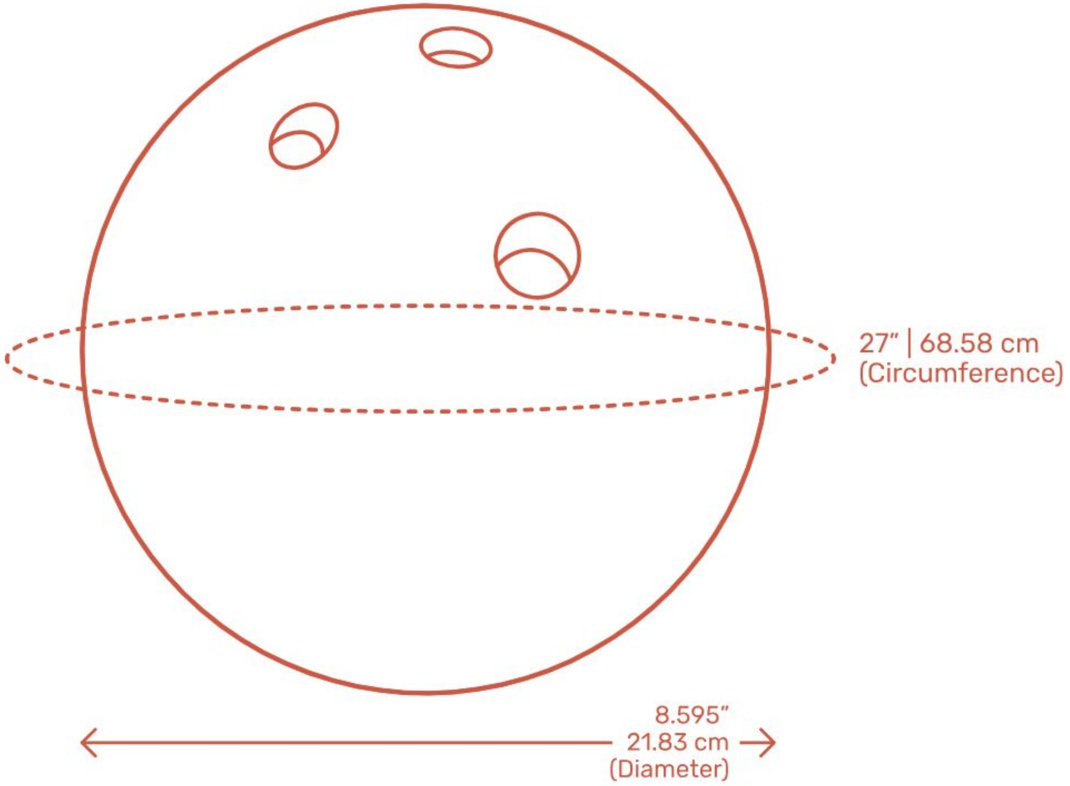
- Compute Homography Leveraging lane boundaries and known coordinates of edges



$$\mathbf{x}_{\text{metric}} = H\,\mathbf{x}_{\text{image}}$$

Lane Rectification and 3D Reconstruction: 3D Trajectory Reconstruction

- Ball contact point with lane is estimated for every frame (interpolated radius)
- True ball radius is known
- 3D position of ball is obtained by projecting image points into rectified plane and fixing Z to true radius



Experimental Activities and Results

- Preprocessing and Edge Detection Comparative Analysis
- Top Boundary Localization via Object-Centric Regression
- Trajectory and Spin Detection Results
- Perspective Rectification and Homography Validation
- Trajectory Reconstruction and 3D Spatial Visualization

Preprocessing and Edge Detection Comparative Analysis

- Comparative analysis to determine best edge detection and single channel conversion method

Table 1: Quantitative success rates of boundary detection for each preprocessing and edge detection combination across 9 video clips.

Grayscale Method	Edge Method	clip1	clip2	clip3	clip5	clip6	clip7	clip8	clip11	clip13	Sum
default	sobel	1	1	1	0	1	1	0	1	0	6
default	canny	0	0	0	0	0	1	0	1	0	2
default	laplacian	0	0	1	0	1	1	0	1	1	5
lightness	sobel	1	1	0	0	0	1	0	1	1	5
lightness	canny	1	0	1	0	0	0	0	1	0	3
lightness	laplacian	0	0	1	0	0	0	0	1	1	3
hue	sobel	0	0	0	0	0	0	0	0	0	0
hue	canny	0	0	0	0	0	0	0	0	0	0
hue	laplacian	0	1	1	0	0	0	0	0	0	2
r_g_minus_b	sobel	1	0	0	1	1	1	0	0	0	4
r_g_minus_b	canny	1	1	1	1	0	1	0	0	0	5
r_g_minus_b	laplacian	1	1	1	0	0	0	0	0	0	3
pca	sobel	1	1	1	1	1	1	0	1	0	7
pca	canny	1	0	0	1	1	0	0	1	0	4
pca	laplacian	0	0	1	0	1	1	0	1	1	5

Top Boundary Localization via Object-Centric Regression

- Template matching + RANSAC robustly identifies the top boundary

Trajectory and Spin Detection Results

Pipeline applied to eight clips across two release sessions

- Session A: clips 9-12
- Session B: clips 13-16

Trajectory: Consistent detection of the ball across frames, even when the ball is occluded.

Trajectory and Spin Detection Results

Spin: per-frame estimates of the rotation axis and angular velocity

$$RPM = \frac{\theta \cdot fps}{2\pi} \cdot 60$$

Session	Clip	Frames	FPS	Avg. θ (rad/frame)	Rev Rate (RPM)
A	clip_9	98	30	0.4420	128.1
	clip_10	125	30	0.4530	129.8
	clip_11	207	30	0.5075	145.4
	clip_12	109	30	0.4831	138.4
B	clip_13	196	30	0.3678	105.4
	clip_14	118	30	0.3637	104.2
	clip_15	104	30	0.3737	107.1
	clip_16	138	30	0.4063	116.4

Avg: 135.4 RPMs

STD: 7.8 RPMs

Avg: 108.3 RPMs

STD: 5.5 RPMs

Trajectory and Spin Detection Results

Final visualization of ball's spin (yellow) and rotation axis (red).

Perspective Rectification and Homography Validation

Figure 15: Bird's-eye view perspective rectification. The computed homography matrices eliminate perspective distortion, rendering the lane boards parallel. The horizontal alignment of the arrows and pins across all trials confirms that true metric proportions are successfully preserved.

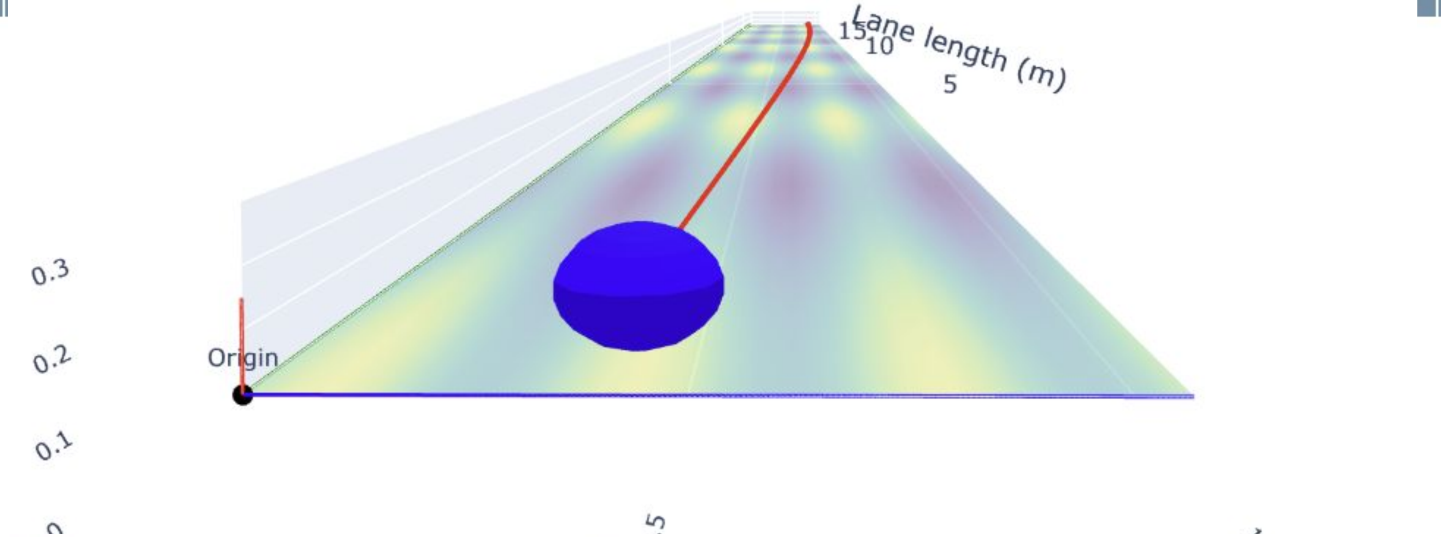
Trajectory Reconstruction and 3D Spatial Visualization

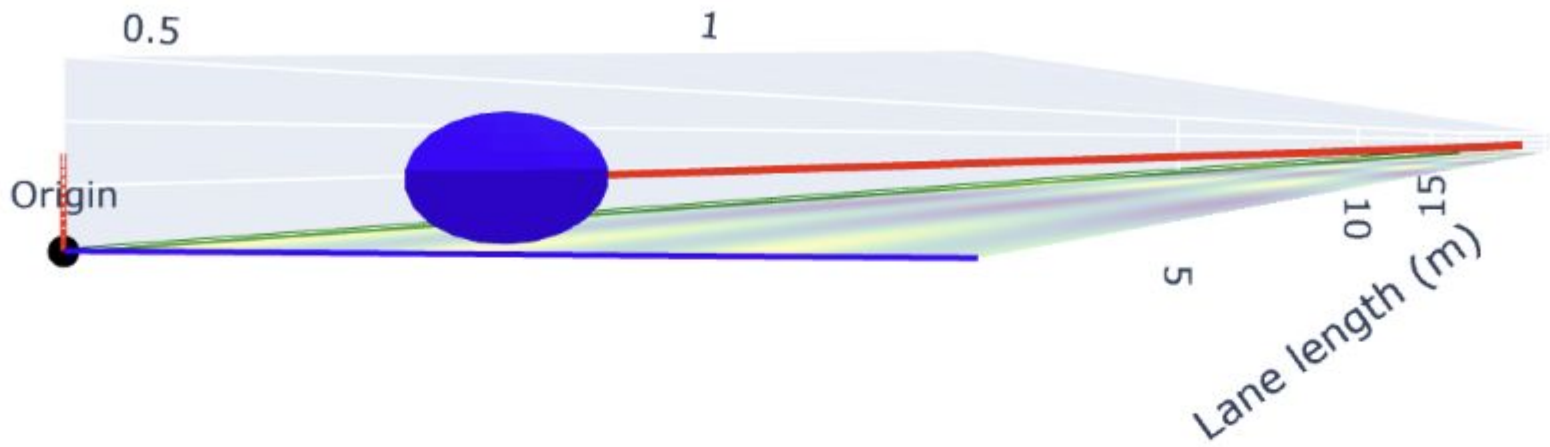
Left: Interpolated tracking data is in red on the original image plane.

Right: Warped bird's-eye visualization to exaggerate hook

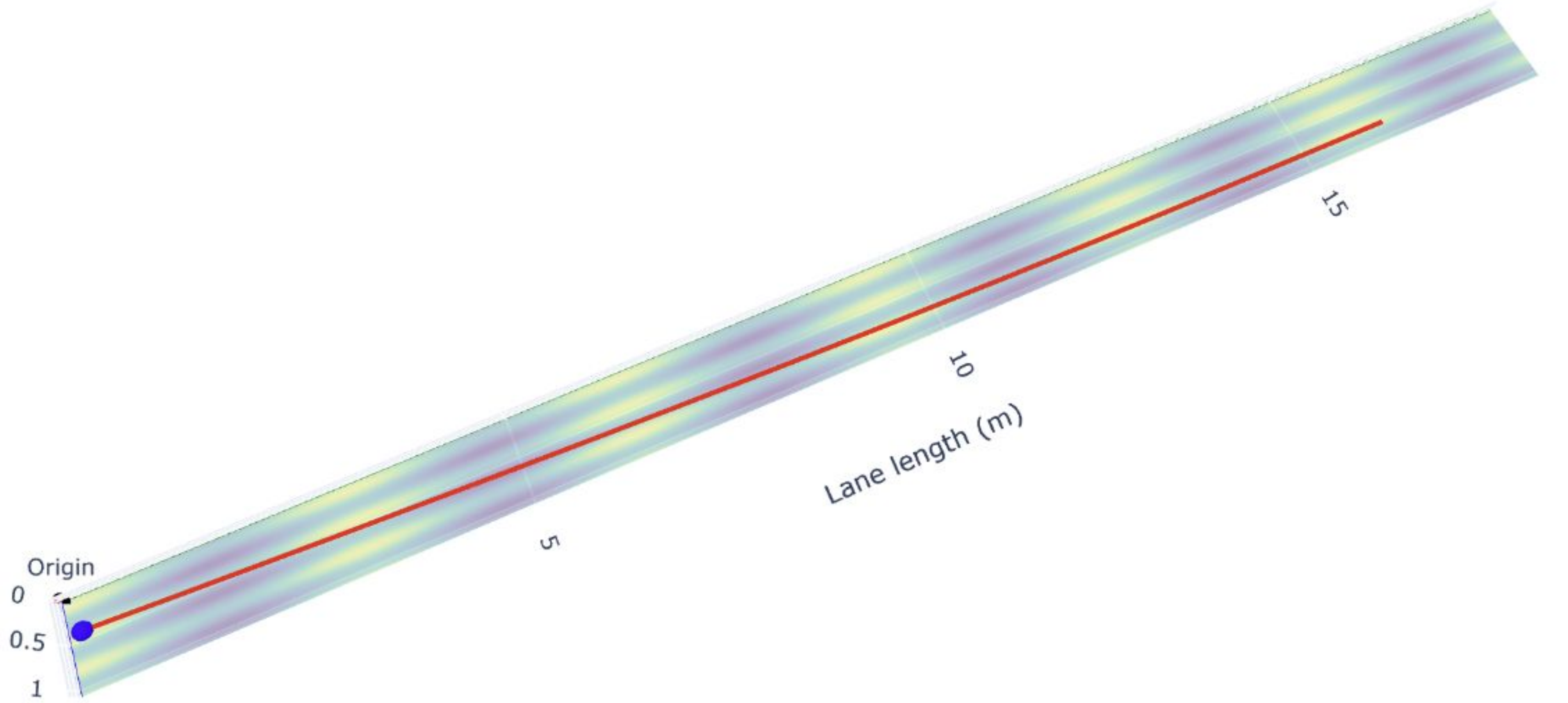
Trajectory Reconstruction and 3D Spatial Visualization

Reconstructed 3D Trajectory Visualization





Trajectory Reconstruction and 3D Spatial Visualization



Conclusions

Developed solution is robust:

- Successful detection of lane in different environments
- Ball tracking correctly tracked ball even under bowler occlusion
- The solution can distinguish between 2 bowlers with different playing styles/level
- Produced a metrically accurate and visually coherent 3D reconstruction of ball’s trajectory

Future Work

- Explore the integration of deep learning for lane detection in severe deviations in composition
- Implement 3D reconstruction without the assumption that the ball is always in contact with lane
- explore frame-by-frame homography estimation or camera tracking algorithms for camera movement